j4_display_left

Jukebox display board for the Johnny 4 robot controller -- the portrait display on the LEFT of the panel (renamed from j4_display when j4_display_right joined the controller). Receives live control data from the main TTGO T-Display controller board over a wired UART link and renders the jukebox UI on a 3.5" colour TFT.

What this board does

- Receives 25 fps binary packets from the TTGO controller over UART (wired, 115200 baud)
- Displays all 7 potentiometer values (volume, eyes, spotlight, arms, neck, jaw)
- Shows the currently selected audio phrase
- Shows all three battery voltages
- Shows ESP-NOW link health (LINKED / NO LINK) between the controller and robot receiver
- Clears the now-playing highlight when a track stops or finishes (acts on empty-phrase packets too)
- Requests the file list with `LIST?` every 2 seconds until it arrives, so a display reboot recovers the list from the controller's cache
- Displays a "Waiting for controller..." message on startup until the first packet arrives
- Sends a `PING` heartbeat once per second so the controller's connection-status screen can show j4_display as connected

Hardware

Component	Details
Microcontroller	SeeedStudio XIAO ESP32S3
Display	3.5" TFT LCD, 480x320, SPI, ILI9488 driver

Display power

Specs from the official product page (Hosyond MSP3520):

Parameter	Value
VCC supply range	3.3V to 5V (onboard regulator on the module PCB)
Logic IO voltage	3.3V TTL
Power consumption	0.62W
Current at 3.3V	~188 mA
Current at 5V	~124 mA

VCC connects to the XIAO's 3V3 pin. The 3.3V regulator handles the display (~188 mA) plus the ESP32S3 fine -- it's rated ~700 mA so there's plenty of headroom. The 5V pin works too and takes some load off the regulator, but it's only live when USB is connected.

The backlight (D0/GPIO1) supports PWM dimming if you need to cut power draw.

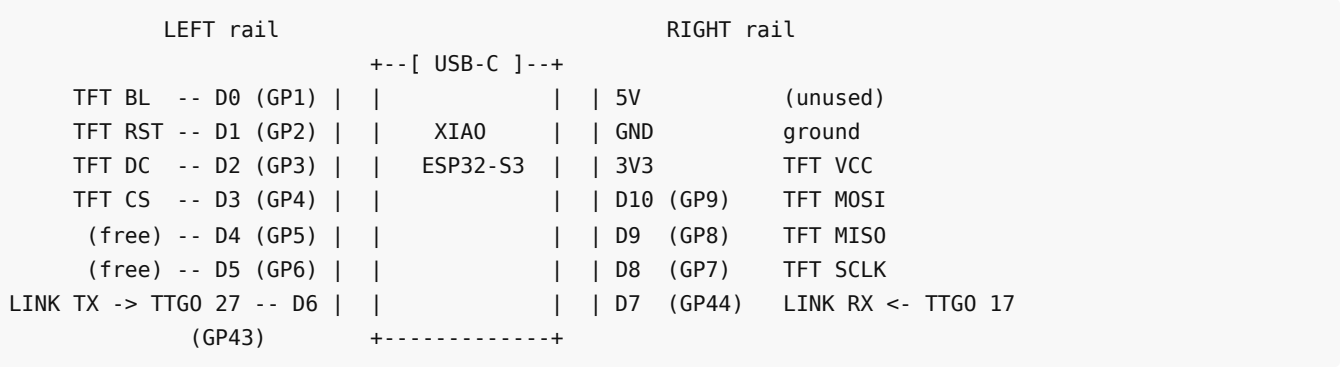
Pin assignments

XIAO Pin	GPIO	Function

3V3	n/a	Display VCC (3.3V power)
GND	n/a	Display GND, shared with TTGO GND
D0	GPIO1	TFT backlight (HIGH = on)
D1	GPIO2	TFT RST
D2	GPIO3	TFT DC
D3	GPIO4	TFT CS
D4	GPIO5	Free (I2C SDA if needed later)
D5	GPIO6	Free (I2C SCL if needed later)
D6	GPIO43	UART TX to TTGO GPIO27
D7	GPIO44	UART RX from TTGO GPIO17
D8	GPIO7	TFT SCLK
D9	GPIO8	TFT MISO
D10	GPIO9	TFT MOSI

Pin diagram

Physical layout, component side up, USB-C at the TOP. The two header rails read top-to-bottom as shown.



The ILI9488 TFT uses BL/RST/DC/CS (D0-D3) plus SCK/MISO/MOSI (D8/D9/D10); D6/D7 are the UART link to the TTGO; D4/D5 are free.

Wiring to TTGO controller

3.3V logic on both sides -- no level shifter needed.

XIAO	TTGO
D7 (GPIO44) RX	GPIO17 TX
D6 (GPIO43) TX	GPIO27 RX
GND	GND

Wiring to ILI9488 display module

XIAO	Display pin
3V3	VCC
GND	GND
D0 (GPIO1)	LED / BL (backlight)
D1 (GPIO2)	RST
D2 (GPIO3)	DC
D3 (GPIO4)	CS
D8 (GPIO7)	SCK
D9 (GPIO8)	MISO (SDO)
D10 (GPIO9)	MOSI (SDI)

UART packet protocol

The TTGO sends a fixed 49-byte packed binary struct (`disp_pkt_t`) at 25 fps:

Bytes	Field	Description
2	`magic`	Frame sync header: 0xAB, 0xCD
1	`volume`	Volume knob (0-100)
1	`eyes`	Eye position (0-255)
1	`spot`	Spotlight intensity (0-255)
1	`left_arm`	Left arm position (0-255)
1	`right_arm`	Right arm position (0-255)
1	`neck`	Neck position (0-255)
1	`jaw`	Jaw position (0-255)
2	`bat1_mv`	Controller battery (millivolts, uint16)
2	`bat2_raw`	Robot battery 2 raw value from receiver
2	`bat3_raw`	Robot battery 3 raw value from receiver
1	`connect_ok`	ESP-NOW link health (1 = ok, 0 = error)
32	`phrase`	Selected audio phrase, null-terminated
1	`checksum`	XOR of all preceding bytes

The receiver re-syncs on the 0xAB 0xCD magic header after any glitch. Packets that fail the checksum are silently dropped.

Dependencies

Managed via PlatformIO (`platformio.ini`):

Library	Version
Bodmer/TFT_eSPI	latest from GitHub

TFT_eSPI is configured entirely through `build_flags` in `platformio.ini` -- no `User_Setup.h` file needed.

Building and uploading

```
# Build
pio run

# Upload
pio run --target upload

# Monitor serial output (USB CDC debug)
pio device monitor
```

If the upload fails, hold the **BOOT** button on the XIAO, tap **RESET**, then release BOOT to force download mode.

Related projects

- **j4_controller** -- the TTGO T-Display transmitter that sends data to this board
- **j4_display_right** -- the landscape display + pot-label board on the right of the panel